Laying the Foundation

A combination of unexpected events and prior planning sent the CFL market soaring in 2001.

by Chris Calwell and John Zugel

It has been 23 years since the compact fluorescent lamp (CFL) was first introduced. Over its first two decades of sales, the CFL made only modest incursions into the residential marketplace thoroughly dominated by a century-old lamp design—the incandescent bulb. Then, in 2001, the market share gains for CFLs outpaced all of the gains achieved in the first 260 months of the products' existence. National sales of CFLs grew from 0.5% share in mid-2000 to 1.6% share in mid-2001, reaching 2.1% by the fourth quarter of 2001. In this same year, California sales absolutely soared, reaching a peak of 8.5% in the second quarter and stabilizing at between 5% and 6% in the third and fourth quarters of 2001 (see Figure 1). What happened?

The events of 2001 built on the foundations laid by energy efficiency programs in previous years. And on top of this groundwork, a number of extraordinary factors came into play in 2001, particularly in California. Any one or two of them alone might have led to a banner year of sales, but together they created something of a "perfect storm"—a confluence of factors at the right time and place to drive an unprecedented market outcome.

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Dramatic Growth

Between 1997 and 1999, a number of things occurred that planted the seeds for dramatic future growth of efficient lighting sales. First, the EPA launched an Energy Star labeling program for residential light fixtures, and later an Energy Star labeling program for screw-based CFLs as well. During this same period, Pacific Northwest National Laboratories (PNNL) accelerated sales of high-quality subcompact (approximately incandescent-sized) CFLs through a targeted procurement and testing effort.

In California, starting in 1999, a utility-led effort, the California Residential Lighting and Appliance Program (CRLAP) undertook a broad, sustained initiative to build a conscious market preference for Energy Star-labeled bulbs, torchieres, and hard-wired fixtures. This included

- a decisive shift from bill stuffers and utility-driven advertising to in-store advertising materials and events, including point-of-purchase banners and shelf-mounted signage, frequent sidewalk sales, and heavily publicized torchiere turn-in events;
- a massive statewide effort to build retailer infrastructure for marketing the products, including professional training of sales representatives and regular visits by field staff to each store to display merchandise attractively, tabulate inventory levels, and maintain point-of-purchase displays; and
- significant investments of incentive dollars into competitively allocated manufacturer buydowns, which maximized price leverage per dollar invested. More importantly, this investment greatly amplified competition among manufacturers. Those who successfully moved their initial allocations of incentive dollars through retail channels by the program deadline could obtain additional allocations from their less successful competitors. This gave upstart manufacturers a foot in the door when competing for retail distribution space with larger, well-established competitors.

Collectively, these program elements helped accomplish something previous California utility programs had not: They built a truly competitive marketplace in efficient lighting. Suddenly, energy-efficient lighting became an important enough revenue source to Home Depot, Lowe's, Home Base, Orchard Supply, and other retailers to cause them to invest their own promotional resources (such as discounts and advertising) to grab market share from their competitors. More importantly, however, the ongoing investment by California throughout the 1990s and 2000 was quietly building an efficiency infrastructure that would pay enormous dividends in 2001, when unprecedented changes in the energy landscape would drive consumers to stores in record numbers to purchase energy-efficient products.

The 2001 Experience

Even allowing for sharp declines in the average selling price of CFLs, CFLs outsold incandescent bulbs on a dollar basis in California for the full year 2001. Here are some of the factors that contributed to this increase in sales.

Episodic power shortages, rolling blackouts, and the fear that they would become more frequent and last longer. As reserve margins of electricity supply became tight in California, utilities and regulators switched from a market transformation mode to a resource acquisition mode. Instead of steadily decreasing incentives and relying primarily on marketing and education to drive sales, the utilities offered rebates that nearly equaled the purchase price of a CFL. The rebate amounts were not particularly high by historical standards. PG&E, for example, offered rebates of $3 per bulb compared to rebates of $4 to $7 per bulb in the early 1990s, yet achieved CFL sales of approximately 7 million units in 2001—a massive increase compared to year 2000 volumes.

Rebates alone could not drive a stampede of purchases. Retail sales staffs were already very familiar with the benefits of CFLs; branding and consumer education messages were firmly in place; and retailers had already established longstanding relationships with numerous competitors. All this made possible the large orders that would follow.

Public awareness messages broadcast by state and city government officials and utilities. These messages urged customers to conserve, specifically by purchasing and using CFLs. The state's Flex Your Power campaign, with initial funding of $20 million and later budget additions, was able to promote simple energy efficiency messages very widely in the state.

Massive direct purchases by state and local governments. The California legislature mobilized $20 million of emergency funding for the California Conservation Corps (CCC) to purchase and distribute free CFLs to nearly half a million low-income households.

Rising electricity rates, especially for usage above baseline amounts. While baseline electric rates held steady, Californians saw very steep increases in their incremental rate for consumption above baseline.

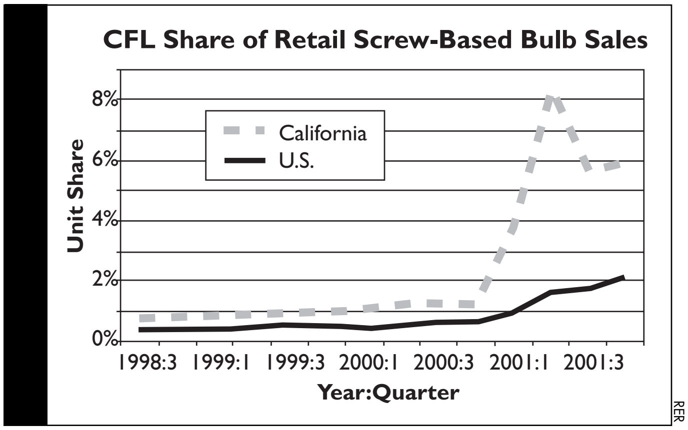

Figure 1. At the same time that the national sales of CFLs grew from 0.5% share in mid-2000 to 1.6% share in mid-2001, California sales absolutely soared, reaching 4% share in the first quarter of 2001 and a peak of 8.5% in the second quarter.

Unprecedented competition among manufacturers and retailers. This competition resulted in substantial internal discounting (manufacturers and retailers accepting lower gross margins than they typically would for CFLs). It created a "virtuous cycle" in which price cuts drove greater sales, increasing economies of scale in manufacturing and making possible even lower prices on subsequent orders.

The imposition of steep tariffs on Asian manufacturers. Tariffs of up to 75% were set by the European Union in response to charges of dumping (below-cost selling). This caused numerous low-cost suppliers from Asia to shift the focus of their marketing efforts to North America, greatly increasing the available supply of products and driving additional price competition. This increase in manufacturers is perhaps most evident in the Energy Star CFL program data. At the beginning of 2001, the program included 17 manufacturers offering 161 qualifying products. By the end of the year, that list had grown to include 94 manufacturers and 455 qualifying products.

Lasting Effects

A number of market changes have occurred that are likely to persist even after utility rebate levels and promotional efforts drop off. Unprecedented competition has fueled a wave of product innovation and aggressive pricing that will make CFLs affordable and potentially attractive to most purchasers from now on. Stores like Home Depot, Ikea, Costco, and Kmart have made inexpensive CFLs a standard promotional item, in many cases bypassing major manufacturers to offer a private-label brand at a lower price.

Not all of these private-label CFLs meet Energy Star requirements, however, which raises legitimate questions about their performance and longevity. Energy Star has gained enough support in the marketplace to be willing to limit participation to products verified to have high quality. Thus the program has moved from a recruitment phase to a discernment phase, requiring manufacturers to meet more stringent specification and testing requirements beginning in July 2002. It no longer needs simply to encourage consumers to buy any CFL instead of an incandescent bulb; it can instead direct consumers toward high-quality CFLs with superior efficiency and performance.

Retailers now clearly understand the benefits of CFL technology, both to consumers (energy savings and convenience) and to the retailers themselves (higher sales revenues per unit of shelf space than incandescent bulbs). As a result, some level of in-store CFL promotion will now be self-sustaining, allowing the utilities to retarget or reduce their funding and still achieve a given level of market success.

Utilities, regional groups, and state governments have come to understand that a largely transformed CFL market represents a very fruitful place to prospect for resource acquisition, both on a short-term and on a long-term basis. The CCC program generated a payback of approximately one year on a $20 million investment of public funds, though long-term evaluations will be needed to verify the persistence of savings and continued usage rates.

Most importantly, the program put approximately 6 times more dollars in the hands of low-income residents through long term energy savings than if it had simply given the $20 million to the residents directly. This multiplier effect is likely to feature prominently in future low-income programs, or in others that need to achieve demand reductions quickly to meet local or regional power shortages.

One overarching lesson from the events of 2001 is that market transformation takes time. CFLs ground their way to begrudging market acceptance for more than two decades before finally gaining traction with consumers. That period reflects multiple generations of revision to the technology itself, the trying and refining of numerous utility program approaches, and a great deal of hard work by a whole array of allies. Large incentives by themselves do not guarantee large energy savings, but they can stimulate a dramatic short-term response in an already substantially transformed, competitive market.

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For more information:

The full report of the CFL market study, "2001: A CFL Odyssey," can be found in *Proceedings of the 2002 ACEEE Summer Study on Energy Efficiency in Buildings*. Washington DC: American Council for an Energy-Efficient Economy, Electronic Library, 2002.